Beyond Null Hypothesis Testing

Evaluating Hypotheses the Way You Intuitively Want

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Problem and possible solutions

Spoiler alert :-)

- Interested in informative, directional hypotheses.
- Only test the null (possibly with post-hoc tests).
- But simple to do better in both Bayesian & non-Bayesian world.

I will show you two methods (nl., bain and GORIC(A)) that will be directly applicable to basically any model you might be using.

ANOVA Example: Comparisons of 3 Means

Examine the difference in happiness
between three types of "treatments":
(1) new treatment, (2) current treatment, and (3) no treatment.

ANOVA

$$y_j = \mu_1 d_{1j} + \mu_2 d_{2j} + \mu_3 d_{2j} + \epsilon_j,$$

where μ_i $(i = 1, 2, 3)$ is the mean in group i,
d_{ij} is a dummy variable denoting group membership,
ϵ_j is an error term and $\epsilon_j \sim N(0, \sigma^2)$.

$(n_i = 20, 20, 31)$

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Hypotheses of interest

Let us say, we expect that

- new treatment (1) works better than the other two (2 & 3): i.e., $\mu_1 > \mu_2$ and $\mu_1 > \mu_3$,
- current treatment (2) works better than no treatment (3): i.e., $\mu_2 > \mu_3$.

This leads to the theory-based hypothesis:

$$H_1: \quad \mu_1 > \mu_2 > \mu_3.$$

Note that '<" denotes "smaller than" and ">" denotes "larger than".

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Testing the null hypothesis

Example 3 means

Often the next step:

Test H_0 with ANOVA F test:

$$\begin{aligned} H_0 : &\quad \mu_1 = \mu_2 = \mu_3, \\ H_a : &\quad \text{not } H_0. \end{aligned}$$

Then, reject or not-reject ('accept') H_0.

But, often not interested in H_0!

Cannot say anything about $H_1 : \mu_1 > \mu_2 > \mu_3$.

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Conclusions w.r.t. hypothesis of interest $H_1 : \mu_1 > \mu_2 > \mu_3$

Example 3 means

With post-hoc test (e.g., Bonferroni) and on basis of the sample means, we could say a bit more.

But, what if

- $\mu_1 \neq \mu_2$, $\mu_1 \neq \mu_3$, $\mu_2 = \mu_3$,
- $\overline{y}_1 > \overline{y}_2$,

which implies $\mu_1 > \mu_2 = \mu_3$.
But also not interested in this.

Furthermore, inconsistent results are possible:
$\mu_1 = \mu_2$, $\mu_2 = \mu_3$, and $\mu_1 \neq \mu_3$,
which is logically impossible, since the first two imply $\mu_1 = \mu_3$.
Using planned contrasts would overcome this, but still = restrictions then.

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Confirmatory methods

Most researchers are able to specify "order-restricted" / "informative" / "theory-based" hypotheses.

Use **knowledge and/or expertise in hypothesis**
(for the methods in this presentation: thus not in a prior).

Confirmation

Compare only prespecified hypotheses
including order restrictions ($<$, $>$, but also $=$).
Limited set.

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Methods to evaluate theory-based hypotheses

- Hypothesis testing: Fbar ($\bar{F}$) test
  (renders p-value and can test only one theory-based hypothesis)
- (Confirmatory) Bayesian model selection (BMS)
- Confirmatory model selection using information criteria:
  GORIC and GORICA

Note: 'model' refers to hypothesis.

Confirmation more power than exploration.

- Kuiper, R. M., and Hoijtink, H. (2010). Comparisons of Means Using Exploratory and Confirmatory Approaches. *Psychological Methods, 15(1)*, 69–86.
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Bayesian model selection using bain

Reference:
Hoijtink, H., Mulder, J., van Lissa, C., and Gu, X. (2018). A tutorial on testing hypotheses using the Bayes factor. *Psychological Methods, 24*, 539-556.

Bayes Factor

Balancing Fit and Complexity

The Bayes factor quantifies the relative support in the data for two hypotheses.

For example, for

$$H_m : \mu_1 > \mu_2 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3$$

Interpretation:
After observing the data, H_m is BF_{mu} times as likely as H_u, with, for example, BF_{mu} is .2, 5, or 10.

The Bayes factor balances fit and complexity; as described next.

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Balancing Fit and Complexity

A (very) loose interpretation of the meaning of fit

$$H_m : \mu_1 > \mu_2 > \mu_3$$

if $\bar{x}_1 = 7$ & $\bar{x}_2 = 4$ & $\bar{x}_3 = 2$ the fit is good

if $\bar{x}_1 = 2$ & $\bar{x}_2 = 4$ & $\bar{x}_3 = 7$ the fit is bad

Bayes Factor

Balancing Fit and Complexity

A (very) loose interpretation of the meaning of complexity

$$H_1 : \mu_1 = \mu_2 = \mu_3$$

very parsimonious, the means have to be exactly equal.

$$H_1 : \mu_1 > \mu_2 > \mu_3$$

one ordering of three means: 1-2-3, thus is parsimonious.

$$H_2 : \mu_1 > (\mu_2, \mu_3)$$

2 orderings of three means: 1-2-3 and 1-3-2, less parsimonious.

$$H_u : \mu_1, \mu_2, \mu_3$$

contains all six possible orderings of three means, not parsimonious.

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Balancing Fit and Complexity

Three forms of hypotheses and Bayes factors involving $H_m : \mu_1 > \mu_2 > \mu_3$

1. BF_{mu} evaluating H_m versus $H_u : \mu_1, \mu_2, \mu_3$;
2. $BF_{mm'}$ evaluating H_m versus $H_{m'}$: a competing hypothesis;
3. BF_{mc} evaluating H_m versus H_c : not H_m, that is, its complement.

H_m is $BF_{m.}$ times as likely as $H_.$.
Demonstrated next via an example.

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Another example

Other example: 3 factors (Condition, Gender Role, and Sex) with corresponding group numbers:

	Condition		
	Masculine	Feminine	Neutral
Masculine Men	1	2	3
Feminine Men	4	5	6
Masculine Women	7	8	9
Feminine Women	10	11	12

Van Well, S., Kolk, A.M., Klugkist, I. (2008). Effects of Sex, Gender Role Identification, and Gender relevance of Two Types of Stressors on Cardiovascular and Subjective Responses: Sex and Gender Match/Mismatch Effects. Behavior Modification, 32, 427 - 449.

Informative Hypotheses

H_1

	Condition		
	Masculine	Feminine	Neutral
Masculine Men	1	2	3
Feminine Men	4	5	6
Masculine Women	7	8	9
Feminine Women	10	11	12

Sex Match Effect

$$H_1 : (\mu_1, \mu_4) > (\mu_2, \mu_3, \mu_5, \mu_6) \text{ and } (\mu_8, \mu_{11}) > (\mu_7, \mu_9, \mu_{10}, \mu_{12})$$

Informative Hypotheses

H_2

	Condition		
	Masculine	Feminine	Neutral
Masculine Men	1	2	3
Feminine Men	4	5	6
Masculine Women	7	8	9
Feminine Women	10	11	12

Gender Role Match Effect

$$H_2 : (\mu_1, \mu_7) > (\mu_2, \mu_3, \mu_8, \mu_9) \text{ and } (\mu_5, \mu_{11}) > (\mu_4, \mu_6, \mu_{10}, \mu_{12})$$

Informative Hypotheses

H_3

	Condition		
	Masculine	Feminine	Neutral
Masculine Men	1	2	3
Feminine Men	4	5	6
Masculine Women	7	8	9
Feminine Women	10	11	12

Sex Mismatch Effect

$$H_3 : (\mu_2, \mu_5) > (\mu_1, \mu_3, \mu_4, \mu_6) \text{ and } (\mu_7, \mu_{10}) > (\mu_8, \mu_9, \mu_{11}, \mu_{12})$$

Informative Hypotheses

H_4

	Condition		
	Masculine	Feminine	Neutral
Masculine Men	1	2	3
Feminine Men	4	5	6
Masculine Women	7	8	9
Feminine Women	10	11	12

Gender Role Mismatch Effect

$$H_4 : (\mu_4, \mu_{10}) > (\mu_5, \mu_6, \mu_{11}, \mu_{12}) \text{ and } (\mu_2, \mu_8) > (\mu_1, \mu_3, \mu_7, \mu_9)$$

Bayes Factor

Interpreting (the Size of) the Bayes Factor

Three forms of comparisons and thus Bayes factors:

1. Select the best of a set of hypotheses using BF_{mu}
2. Compare two competing hypotheses using $BF_{mm'} = BF_{mu}/BF_{m'u}$
3. Compare "my theory" with "not my theory" using BF_{mc}

	f_m	c_m	BF_{mu}	BF_{mc}
H_1 : Sex Match	.0039	.012	.32	.32
H_2 : Gender Role Match	.0725	.012	5.85	6.44
H_3 : Sex Mismatch	.0007	.012	.06	.06
H_4 : Gender Role Mismatch	.0001	.012	.01	.01

H_m is $BF_{m.}$ times as likely as $H_.$.

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PMPs

PMPs Example

Posterior Model Probabilities (PMPs) quantify the support in the data for each hypothesis.

This is a re-scaling of the BFs such that they sum to 1
(for the hypothesis set of interest).

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.04	1/5
H_2 : Gender Role Match	.0725	.012	5.85	.81	1/5
H_3 : Sex Mismatch	.0007	.012	.06	.01	1/5
H_4 : Gender Role Mismatch	.0001	.012	.01	.00	1/5
H_u :				.14	1/5

where PRI_m denotes the *prior* probability, that is, the (subjective) support / likeliness for the hypothesis *before* observing the data.

Bayes Factor

Interpreting (the Size of) the Bayes Factor & PMPs

1. The Bayes factor **is** a measure of support (also for the null-hypothesis H_0, if(!) H_0 is of interest).
2. "Something is going on and **we (often) do know what!**".
3. Note that the Bayes factor **can be indecisive**: A value around 1 denotes "the data don't tell us which hypothesis to prefer, they are equally likely".
4. One **can compare** more than two (informative) hypotheses.
5. The Bayes factor **selects the best of the hypotheses under consideration**. Hence, make sure that not all hypotheses are "wrong"; hence, include a failsafe (discussed later).

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Interpreting (the Size of) the Bayes Factor & PMPs

When is the Bayes factor large enough?

1. Guidelines by Jeffreys (1969) and Kass and Raftery (1995),
   but this is determined for a null hypothesis versus its complement
   (i.e., the unconstrained hypothesis);
   not for informative hypotheses versus another one.
2. 'Genreal' cut-offs will lead to a return of sloppy science and publication bias (when no pre-registration or pre-registered report).

Subjectivity of Bayesian Hypotheses Evaluation

1. Which hypotheses to evaluate?
2. How to formalize hypotheses?
   E.g.: $(\mu_1, \mu_2) > (\mu_3, \mu_4)$ or $\mu_1 = \mu_2 > \mu_3 = \mu_4$
3. The (implicit) choice for equal prior model probabilities.
4. The specification of the prior distribution.
   In bain, uninformative; but: choice matters if "$=$"-restriction.

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Information criteria (ICs)

An IC, like GORIC, also balances fit and complexity:

Describe data as good as possible (fit)
with fewest number of parameters (simplicity / non-complexity).

GORIC: Lowest value is best

GORIC is like AIC expected distance from the truth (KL-distance).
One wants smallest distance.

Or: One wants the smallest 'misfit + complexity'.
Note: GORIC = -2 fit + 2 complexity = 2 misfit + 2 complexity

Hence, smallest value is best.

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Let's return to first ANOVA Example

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Theory-based hypothesis:

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where ">" denotes "larger than".

Possible competing hypothesis:

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Example GORIC value

Based on Palmer & Gough (2007) Data

$$
\begin{aligned}
H_1 &: \quad \mu_1 > \mu_2 > \mu_3, \\
H_2 &: \quad \mu_2 > \mu_1 > \mu_3, \\
H_u &: \quad \mu_1, \ \mu_2, \ \mu_3.
\end{aligned}
$$

GORIC

Model	Fit	Complexity	GORIC
H_1	-191.89	2.81	389.41
H_2	-193.70	2.81	393.03
H_u	-191.89	4.00	391.79

H_1 has lowest GORIC:
has best balance between fit and complexity
(and is not weak).

Example GORIC value

Based on Palmer & Gough (2007) Data

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\begin{aligned}
H_1 &: \quad \mu_1 > \mu_2 > \mu_3, \\
H_2 &: \quad \mu_2 > \mu_1 > \mu_3, \\
H_u &: \quad \mu_1,\ \mu_2,\ \mu_3.
\end{aligned}
$$

GORIC

Model	Fit	Complexity	GORIC
H_1	-191.89	2.81	389.41
H_2	-193.70	2.81	393.03
H_u	-191.89	4.00	391.79

H_1 has lowest GORIC:
has best balance between fit and complexity
(and is not weak).

Interpretation: GORIC weights

GORIC values

GORIC values cannot be interpreted, only compared:
Smallest is best.

GORIC weights (w_m)

w_m quantifies relative support for H_m versus others in the set.
Values between 0 and 1, and they sum to 1.
The bigger, the better.

GORIC weight ratios ($w_m/w_{m'}$)

$w_m/w_{m'}$ quantifies relative support of H_m vs $H_{m'}$.
Values between 0 and infinity:
* If < 1, lack of support for H_m (nl. $w_{m'}/w_m$, support for $H_{m'}$).
* If > 1, support for H_m.

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Example GORIC weights (w_m)

Palmer & Gough (2007) Data

$$\begin{aligned} H_1 &: \quad \mu_1 > \mu_2 > \mu_3, \\ H_2 &: \quad \mu_2 > \mu_1 > \mu_3, \\ H_u &: \quad \mu_1, \ \mu_2, \ \mu_3. \end{aligned}$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 is 0.68 / 0.11 $\approx$ 6.11 times more supported than competing hypothesis H_2.

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Include "unconstrained" hypothesis

If set of hypotheses does not contain a reasonable/good one:
Select the best of set of weak hypotheses.
E.g.: $w_1 = .8$ and $w_2 = .2$.

Prevent choosing a weak hypothesis

Include unconstrained hypothesis H_u (highest fit but also most complex).
E.g.: $w_1 = .08$, $w_2 = .02$, and $w_u = .90$.

If at least one informative hypothesis not weak ($w_1 > w_u$ or $w_1 / w_u > 1$), then (and only then) compare informative hypotheses.

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Example Unconstrained

Palmer & Gough (2007) Data

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

If at least one informative hypothesis not weak ($w_1 > w_u$ or $w_1/w_u > 1$), then compare informative hypotheses.

Hence: H_u is only a failsafe not another hypothesis of interest.

Download 'Guidelines_output_GORIC.html' from `https://github.com/rebeccakuiper/Tutorials`.
On github site, go to Code (green button) and download zip.

Alternative safeguard: Complement of H_m

Alternatively (in case of one hypothesis of interest)

Evaluate hypothesis of interest against its complement;
that is, all other possible hypotheses.

More powerful than against the unconstrained
if H_m has maximum fit.

Reference:
Vanbrabant, L., Van Loey, N., and Kuiper, R. M. (2020). Evaluating a Theory-Based Hypothesis Against Its Complement Using an AIC-Type Information Criterion With an Application to Facial Burn Injury. Psychological Methods, 25(2), 129-142.
https://doi.org/10.1037/met0000238

Failsafe hypotheses: H_c vs H_u

H_m contains 1 ordering of means:

1. $\mu_1 > \mu_2 > \mu_3$

H_c contains 5 orderings of means:

2. $\mu_1 > \mu_3 > \mu_2$
3. $\mu_2 > \mu_1 > \mu_3$
4. $\mu_2 > \mu_3 > \mu_1$
5. $\mu_3 > \mu_1 > \mu_2$
6. $\mu_3 > \mu_2 > \mu_1$

H_u combines H_m and H_c.

Failsafe hypotheses: H_c vs H_u

H_m contains 1 ordering of means:

1. $\mu_1 > \mu_2 > \mu_3$

H_c contains 5 orderings of means:

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4. $\mu_2 > \mu_3 > \mu_1$
5. $\mu_3 > \mu_1 > \mu_2$
6. $\mu_3 > \mu_2 > \mu_1$

H_u combines H_m and H_c.

Example H_1 vs H_c

Palmer & Gough (2007) Data

$$H_1: \quad \mu_1 > \mu_2 > \mu_3,$$
$$H_c: \quad \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H1	-191.89	2.81	389.41	0.79
complement	-192.34	3.69	392.05	0.21

H_1 is 0.79 / 0.21 $\approx$ 3.73 times more supported than its complement H_c, that is, any other hypothesis.

ANOVA Example alternative

Comparisons of 3 Means with minimal/meaningful differences

Examine the difference in happiness
between three types of "treatments":
(1) new treatment, (2) current treatment, and (3) no treatment.

Theory-based hypothesis using minimal or meaningful differences:

$$H_1: \quad \mu_1 - \mu_2 > 0.1 \quad \& \quad \mu_2 - \mu_3 > 0.2,$$

where ">" denotes "larger than".

Example GORIC minimal/meaningful difference

Based on Palmer & Gough (2007) Data

$$\begin{aligned} H_1 &: \quad \mu_1 - \mu_2 > 0.1 \quad \& \quad \mu_2 - \mu_3 > 0.2, \\ H_c &: \quad \text{not } H_1. \end{aligned}$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.76
H_c	-192.18	3.69	391.72	0.24

H_1 is 0.76 / 0.24 ≈ 3.17 times more supported than its complement H_c, that is, any other hypothesis.

Example GORIC minimal/meaningful difference

Based on Palmer & Gough (2007) Data

$$H_1: \quad \mu_1 - \mu_2 > 0.1 \quad \& \quad \mu_2 - \mu_3 > 0.2,$$
$$H_c: \quad \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.76
H_c	-192.18	3.69	391.72	0.24

H_1 is 0.76 / 0.24 $\approx$ 3.17 times more supported than its complement H_c, that is, any other hypothesis.

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GORICA

Similarities with GORIC

- Form: $GORICA_m = -2\ fit + 2\ complexity$.
- Broad type of restrictions.

GORICA: All statistical models

Therefore, GORICA: asymptotic expression for GORIC.
Can be used for all types of statistical models.

Reference:
Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods, 26(5)*, 599–621.
https://doi.org/10.1037/met0000406

Example GORICA

Palmer & Gough (2007) Data

$$H_1: \quad \mu_1 > \mu_2 > \mu_3,$$
$$H_c: \quad \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORICA	GORICA weights
H1	-1.90	1.81	7.42	0.79
complement	-2.34	2.69	10.06	0.21

H_1 is 0.79 / 0.21 $\approx$ 3.74 times more supported than its complement H_c, that is, any other hypothesis.

Note: GORIC weights are the same.

Some literature

- Logistic Regression Modeling
  - Article: https://doi.org/10.1037/met0000406
- GORICA on SEM
  - Article:
  https://www.tandfonline.com/doi/full/10.1080/10705511.2020.1836967.
  - R scripts: https://github.com/rebeccakuiper/GORICA_in_SEM.
- GORICA on cross-lagged panel model (CLPM) - Article:
  https://doi.org/10.1111/bjep.12455.
  - R scripts: https://github.com/rebeccakuiper/GORICA_in_SEM.
- GORICA on Random-Intercept CLPM (RI-CLPM)
  - Article: Chuenjai Sukpan and Rebecca M. Kuiper (submitted 2023). How to evaluate causal dominance hypotheses in lagged effects models.
  - R scripts: https://github.com/Chuenjai/Causal-dominance.
- GORICA on CTmeta
  - Article: https://doi.org/10.1080/10705511.2020.1823228.
  - R scripts: https://github.com/rebeccakuiper/GORICA_on_CTmeta.
- GORICA on Meta-analysis
  - Article: https://doi.org/10.3390/e24111525.
  - R scripts: https://github.com/rebeccakuiper/GORICA_on_MetaAn.

Note: On github site, go to Code (green button) and download zip.

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Possibilities multiple studies (1/2)

- Update hypotheses.

  First data set (or a part of it) generates one or more hypotheses.
  Other data set (or part) used to determine evidence / support.

  - html tutorial: `https://github.com/rebeccakuiper/Tutorials/blob/main/Tutorial_GORIC_restriktor_UpdateHypo.html`
  - R script tutorial: `https://github.com/rebeccakuiper/Tutorials/blob/main/Hands-on%20files/Hands-on_4_GORIC_UpdateHypo_restriktor.R`

  Note: On github site, go to Code (green button) and download zip.

Possibilities multiple studies (2/2)

- Aggregate evidence for hypotheses.

  Aggregate the support for theories (diverse designs allowed).

  Bear in mind: Meta-analysis aggregates parameter estimates or effect sizes which need to be comparable (often same designs required).

  - html tutorial: https://github.com/rebeccakuiper/Tutorials/blob/main/Tutorial_GORIC_restriktor_AggrSupport.html
  - R script tutorial: https://github.com/rebeccakuiper/Tutorials/blob/main/Hands-on%20files/Hands-on_5_GORICA_CombEv_restriktor.R

  Note: On github site, go to Code (green button) and download zip.

GORIC(A) for Multiple Studies: Aggregating support (= evidence synthesis)

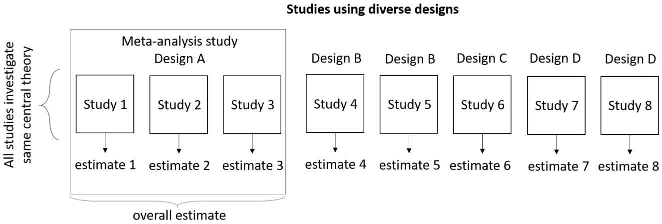
References:

- Kuiper, R.M., Buskens, V.W., Raub, W., and Hoijtink, H. (2013). Combining statistical evidence from several studies: A method using Bayesian updating and an example from research on trust problems in social and economic exchange. *Sociological Methods and Research*, 42 (1), (pp. 60-81) (22 p.).

- Van Lissa, C.J., Clapper, E.-B., and Kuiper, R.M. (submitted 2023). Aggregating evidence from conceptual replication studies using the product Bayes factor. `10.31234/osf.io/nvqpw`

- Kuiper, R.M., and Clapper, E.-B. (to be submitted in 2023). GORIC Evidence Aggregation: Combining Statistical Evidence for a Central Theory from Diverse Studies using an AIC-type Criterion. `https://psyarxiv.com/qv76x/`

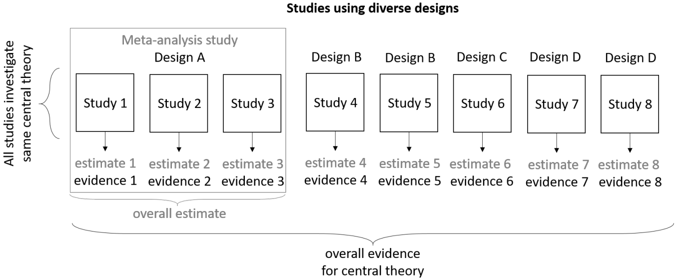
Current best practice



Current best practice is meta-analysis and Bayesian updating.

- Not applicable for diverse research designs.
- Not applicable for incomparable estimates.

Need for new methodology: Evidence Synthesis



Note: All studies do investigate the same, central theory (using diverse designs).

Trust Example: Meta-Analysis versus Evidence Synthesis

Study	Type of model
1	univariate regression
2	univariate regression
3	probit regression
4	three-level logistic regression
	Same design? e.g., same set of predictors?

Conceptual replications!

	Meta-Analysis	Evidence Synthesis
Effect size not required		✓
Deal with diverse designs		✓
Main results	Estimate of effect size	Evidence for hypotheses
Check:		same theoretical relationships?

Reference:

Kuiper, R.M., Buskens, V.W., Raub, W., and Hoijtink, H. (2013). Combining statistical evidence from several studies: A method using Bayesian updating and an example from research on trust problems in social and economic exchange. *Sociological Methods and Research*, 42 (1), (pp. 60-81) (22 p.).

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Example: 4 studies regarding one concept

Study	Type of study	Number of observations n	Type of model
1	survey	895 transactions	univariate regression
2	experiment	348 decisions by 40 subjects	univariate regression
3	experiment	1249 decisions by 125 subjects	probit regression
4	experiment	2160 decisions by 144 subjects	three-level logistic regression
Study	**Outcome y (trust)**		**scale y**
1	effort invested in management		ratio
2	effort invested in management		ratio
3	choice of vignettes		dummy
4	trustfulness		dummy
Study	**Predictor x_1 (past / previous experience)**		**scale x_1**
1	existence relationship with supplier		dummy
2	type of relationship with supplier		interval
3	bought a car from The Autoshop before		dummy
4	number of times a trustee honored trust in the past		ratio
Study	**some of the other predictors**		
1	transaction characteristics, expected future transactions, network embeddedness		
2	transaction characteristics, expected future transactions, network embeddedness		
3	expected future transactions, network embeddedness		
4	future interactions, network embeddedness		

One-Parameter Example: Hypotheses of interest

Main central theory

Previous experience has a positive effect on trust.
For simplicity, only one relationship here, could have been more.

Study-specific hypothesis

$\beta_1 > 0$
Here, for each study the same hypothesis..

Set of central theories

$$H_0 : no\ effect,$$
$$H_> : positive\ effect,$$
$$H_< : negative\ effect.$$

Note 1: Central hypotheses for all studies, not w.r.t. average parameter.
Note 2: In practice, I would not include H_0...

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Note 1: Central hypotheses for all studies, not w.r.t. average parameter.
Note 2: In practice, I would not include H_0...

Example: Trust (y) & previous experience (x_1)

Not full data set (and probit regression), so use

- GORICA (not GORIC) using *goric* function in R package *restriktor*

Input:

- parameter estimates and their covariance matrix

t	$\hat{\beta}_1$	$\hat{\sigma}_{\beta_1}$
1	0.090	0.029
2	0.140	0.054
3	1.090	0.093
4	1.781	0.179

Note: Here, one parameter (β_1); thus, cov. matrix $\hat{\beta}_1$ = variance $\hat{\beta}_1 = \hat{\sigma}^2_{\beta_1}$ (not $\hat{\sigma}_{\beta_1}$)

One-Parameter Example: results per study

using GORICA

Results per study (not aggregated yet)!

Table: GORICA weights ($w_{t,m}$) for Hypothesis H_m in Study t

	$w_{t,m}$			
m / t	1	2	3	4
0	0.013	0.052	0.000	0.000
$>$	**0.979**	**0.916**	**1.000**	**1.000**
$<$	0.008	0.032	0.000	0.000

Note: Weight is at max 1.

So, now on forehand already clear.... but no quantification yet.

One-Parameter Example: Results & Conclusions

using GORICA

Table: Overall GORICA weights ($w^1_{t,m}$) for Hypothesis H_m in Study t

	$w^1_{t,m}$			
m / t	1	2	3	4
0	0.013	0.001	0.000	0.000
$>$	**0.979**	**0.999**	**1.000**	**1.000**
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- $w^1_{4,>} = 1$ => full support for $H_>$
  $w^1_{4,0} = w^1_{4,<} = 0$ => no support for H_0 and $H_<$
- Support for $H_>$ ($w^1_{4,1}$) is highest: favor $H_>$ over H_0 and $H_<$.

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$<$	0.008	0.000	0.000	0.000

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- Support for $H_>$ ($w^1_{4,1}$) is highest: favor $H_>$ over H_0 and $H_<$.

Possible types of sets of studies

- Conceptual replications of same authors, done as a robustness check.
- Searching for direct and indirect/conceptual replications in the literature.
- Using multiple $N = 1$ studies.
- Using different cohorts, where one can measure the variables in their own way.
- Using different subpopulations, possibly using different operationalisations.
- ...

Possible type of sets of studies (1/5)

- Conceptual replications of same authors, done as a robustness check.

  This was done in the Trust example of Buskens and Raub.

  Note: There, the central hypotheses regard one parameter of interest (in each study), but one can compare (absolute values of) multiple parameters or multiple effect sizes.

  For some examples, download 'Tutorial_GORIC_restriktor_evSyn.html' from `https://github.com/rebeccakuiper/Tutorials`.

  Note: On github site, go to Code (green button) and download zip.

Possible type of sets of studies (2/5)

- Searching for direct and indirect/conceptual replications in the literature.

  E.g., using the central hypothesis that the absolute strength of the relationship of communication competence (C) with willingness to communicate in a second language (WTC) is greater than the absolute strength of the relation of communication anxiety (A) with WTC, which is greater than the absolute strength of the relation of motivation (M) with WTC, that is, $|C| > |A| > |M|$;
  where C, A, and M are operationalized differently in the studies.

  - 'Article': Example in bachelor thesis of Martijn Sips
  - R scripts: `https://github.com/rebeccakuiper/Tutorials/tree/main/Examples%20evSyn/Example%20WtC`

  Note: On github site, go to Code (green button) and download zip.

Possible type of sets of studies (3/5)

- Using multiple $N = 1$ studies.

  - Article: Klaassen et al. (2018).
  All for one or some for all? Evaluating informative hypotheses using multiple $N = 1$ studies. *Behavior Research Methods*. 50, 2276–2291.
  `https://link.springer.com/article/10.3758/s13428-017-0992-5`.

Possible type of sets of studies (4/5)

- Using different cohorts, where one can measure the variables in their own way.

  So, possibly using different operationalisation of the same constructs.

  - Article: Zondervan-Zwijnenburg et al. (2020).
  Parental Age and Offspring Childhood Mental Health: A Multi-Cohort, Population-Based Investigation. *Child Development*. 91(3), 964–982.

Possible type of sets of studies (5/5)

- Using different subpopulations, possibly using different operationalisations.

  E.g., the Municipal Health Services (Dutch acronym: GGD) studied the positive consequences of corona on loneliness (a), mental health (b), and stress (b); conditional on sex, age, and health.
  Central hypothesis: $a < b < c$.

  - Article: in progress
  - R scripts: `https://github.com/rebeccakuiper/Tutorials/tree/main/Examples%20evSyn/Example%20corona%20GGD`

  Note: On github site, go to Code (green button) and download zip.

Table of Contents

Software

- GORIC(A)
  - R function *goric* in R package *restriktor*
  - JASP
- GORIC(A) evidence synthesis
  - R function *evSyn* in R package *restriktor*
  - Interactive web application (Shiny app) available from my site (see below).

Websites

https://github.com/rebeccakuiper/Tutorials
www.uu.nl/staff/RMKuiper/Software
www.uu.nl/staff/RMKuiper/Extra2
informative-hypotheses.sites.uu.nl/software/goric/

The End

Thanks for listening!

Are there any questions?

Websites

https://github.com/rebeccakuiper/Tutorials
www.uu.nl/staff/RMKuiper/Software
www.uu.nl/staff/RMKuiper/Extra2
informative-hypotheses.sites.uu.nl/software/goric/

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Generalized Order-Restricted Information Criterion

GORIC

'IC' = -2 fit + 2 complexity

Fit = Maximized order-restricted log likelihood

Maximized log likelihood based on parameters in agreement with H_m.

Complexity = Penalty

Represents: Expected number of distinct parameters.
Here, expected number of distinct mean values plus 1 (because of the unknown variance term).
Details: Function of level probabilities.

Idea complexity

loose interpretation

$$H_1 : \mu_1 > \mu_2 > \mu_3$$

contains 1 ordering of three means, 1-2-3.
Thus, not complex (i.e., parsimonious).

$$H_2 : \mu_1 > \mu_2, \mu_3$$

contains 2 orderings of three means: 1-2-3 and 1-3-2.
Thus, more complex (less parsimonious).

$$H_u : \mu_1, \mu_2, \mu_3$$

contains all six possible orderings of three means.
Thus, is most complex one (least parsimonious).

Example GORIC

Palmer & Gough (2007) Data

$$
\begin{aligned}
H_0 &: \quad \mu_1 = \mu_2 = \mu_3,\\
H_1 &: \quad \mu_1 > \mu_2 > \mu_3,\\
H_2 &: \quad \mu_1 > \mu_2 < \mu_3,\\
H_3 &: \quad \mu_1 < \mu_2 < \mu_3,\\
H_u &: \quad \mu_1,\ \mu_2,\ \mu_3.
\end{aligned}
$$

GORIC

Model	Fit	Complexity	GORIC
H_0	-196.36	2.00	396.71
H_1	-191.89	2.81	389.41
H_2	-192.34	3.19	391.05
H_3	-196.36	2.81	398.34
H_u	-191.89	4.00	391.79

GORIC

$IC_m = -2\ fit_m + 2\ complexity_m$

Broad type of restrictions

More or less: any linear restriction.
e.g., the interaction $H_1 : \mu_1 - \mu_2 < \mu_3 - \mu_4$.

Note

If no inequalities ($<$ and/or $>$), then (G)ORIC = AIC.

References:

- Kuiper, R.M., Hoijtink, H. and Silvapulle, M.J. (2011). An Akaike type information criterion for model selection under inequality constraints. *Biometrika, 98*, 495-501.
- Kuiper, R. M., Klugkist, I., and Hoijtink, H. (2010). A Fortran 90 Program for Confirmatory Analysis of Variance. *Journal of Statistical Software, 34(8)*, 1–31.

GORICA

GORIC: Normal linear models

GORIC can easily be applied to normal linear models
(e.g., ANOVA models or regression models).

GORIC: Other statistical models

In case of other statistical models (e.g., a SEM model), more cumbersome to calculate maximized order-restricted log likelihood and thus GORIC.

GORICA: All statistical models

Therefore, GORICA: asymptotic expression for GORIC.
Can be used for all types of statistical models.

Reference:
Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods, 26(5)*, 599–621.
https://doi.org/10.1037/met0000406

GORICA

Similarities with GORIC

- Form: $GORICA_m = -2\ fit + 2\ complexity$.
- Broad type of restrictions.

Differences compared to GORIC

- Uses asymptotic expression of the likelihood (is a normal): can therefore be easily applied to all types of statistical models. Disadvantage: might work less well in case of small samples.
- Does not need data set; mle's and their covariance matrix suffice.
- Can leave out nuisance parameters (i.e., not part of hypotheses).

Note

In case of normal linear models and/or not too small samples:
GORICA weights = GORIC weights.

Note on BF as an IC

The fit in GORICA refers to the maximum log likelihood.
The fit in BF is related to the maximum likelihood.

Notably, BF can be written as an IC (and vice versa):

$$\begin{aligned}
-2\ log\ BF_{1u} &= -2\ log\ \frac{f_1}{c_1} \\
&= -2\ log\ f_1 \ + \ 2\ log\ c_1 \\
&= -2\ log\ \text{fit}\ H_1 \ + \ 2\ log\ \text{complexity}\ H_1.
\end{aligned}$$

Note: complexity value in BF depends on prior.

Note on GORIC weights vs BF and PMPs

ratio GORIC weights ($w_m/w_{m'}$) $\sim$ Bayes factor ($BF_{mm'}$).

GORIC weight (w_m) $\sim$ posterior model probability (PMP).

1 - w_m = conditional error probability.
Like PMP, w_m depends on set of hypotheses.

Note on hypotheses

1. Only include hypotheses with sound theoretical and/or empirical basis.
   Often a null hypothesis is not of interest.
2. Keep the number of hypotheses included as small as possible.
3. This is a subjective endeavor, aim for inter-peer / inter-subjective agreement.

The Traditional Null Hypothesis

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Cohen (1994) "The Earth is Round p $<$ .05"

Royal (1997) "A power analysis should render $N = 0$"

Only use the null-hypothesis if it is a plausible representation of population of interest

1. Cohen, J. (1994). The earth is round, p¡.05. *American Psychologist, 49,* 997-1003.
2. Royal, R. (1997). *Statistical Evidence. A Likelihood Paradigm.* New York: Chapman and Hall/CRC.

Informative Hypothesis

ANOVA: possible interaction

What is the relation between "knowledge of numbers after watching Sesame Street for a year" and sex & setting.

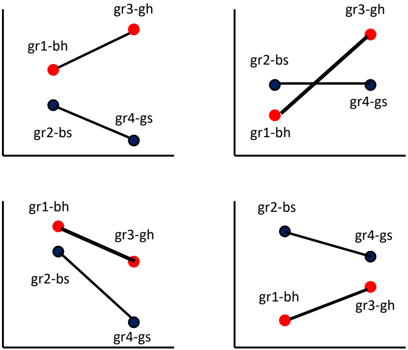
Dependent variable: Knowledge of numbers.

Factors: sex (boy, girl) and setting (watching at home, watching at school).

Gr: 1 = boyhome, 2 = boyschool, 3 = girlhome, 4 = girlschool.

Informative Hypotheses

Example 2: ANOVA Interaction Effect



Interaction bottom right:

$$H_m: \quad gr2 - gr1 > gr4 - gr3 \quad \& \quad gr2 > gr1 \quad \& \quad gr2 > gr4$$

Informative Hypotheses

Example Repeated Measures

Development of depression

	Measurement			
	8 years	12 years	16 years	20 years
Men	μ_1	μ_2	μ_3	μ_4
Women	μ_5	μ_6	μ_7	μ_8

$$H_1 : \mu_5 - \mu_1 > \mu_6 - \mu_2 > \mu_7 - \mu_3 < \mu_8 - \mu_4$$

$$H_2 : \mu_6 - \mu_5 < \mu_7 - \mu_6 > \mu_8 - \mu_7$$

Informative Hypotheses

Example Multiple Regression

$$\text{postnumb}_i = \beta_0 + \beta_1 \times \text{age}_i + \beta_2 \times \text{prenumb}_i + \epsilon_i$$

$$H_1 : \beta_1 > 0, \beta_2 > 0, \beta_1 < \beta_2$$

Note: β_1 and β_2 are only comparable if age and prenumb are standardized

Note on comparable estimates

Until now: comparing means.

Continuous predictors

If compare relative strength/importance of parameters (e.g., $\beta_1 > \beta_2$),
then make sure comparable:
e.g., standardize continuous predictors.

Multiple outcomes

If compare parameters across outcomes,
then (also) standardize outcomes.

Note on type of hypotheses

Absolute strength

Compare strength/importance of parameters: $\beta_1 > \beta_2$.
Compare absolute strength/importance of parameters:
$abs(\beta_1) > abs(\beta_2)$.

About-equality

Equality: $\beta_1 = \beta_2$.
About-equality: $-0.01 < \beta_1 - \beta_2 < 0.01$.

Minimum difference

Difference: $abs(\beta_1) > abs(\beta_2)$.
Minimum difference: $abs(\beta_1) - abs(\beta_2) > 0.01$.

Btw abs() is only possible in goric(), not in bain().

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PMPs

PMPs Example

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.04	$1/5$
H_2 : Gender Role Match	.0725	.012	5.85	.81	$1/5$
H_3 : Sex Mismatch	.0007	.012	.06	.01	$1/5$
H_4 : Gender Role Mismatch	.0001	.012	.01	.00	$1/5$
H_u :				.14	$1/5$

where PRI_m denotes the *prior* probability, that is, an evaluation of the support for the hypothesis *before* observing the data.

Note on failsafe hypotheses (1/2)

H_m contains 1 ordering of means:

1. $\mu_1 > \mu_2 > \mu_3$

H_c contains 5 orderings of means:

2. $\mu_1 > \mu_3 > \mu_2$
3. $\mu_2 > \mu_1 > \mu_3$
4. $\mu_2 > \mu_3 > \mu_1$
5. $\mu_3 > \mu_1 > \mu_2$
6. $\mu_3 > \mu_2 > \mu_1$

H_u combines H_m and H_c.

Note on failsafe hypotheses (2/2)

H_1 contains 1 ordering of means:

1. $\mu_1 > \mu_2 > \mu_3$

H_2 contains 1 ordering of means:

2. $\mu_1 > \mu_3 > \mu_2$

H_c (of set!) contains 4 orderings of means:

3. $\mu_2 > \mu_1 > \mu_3$
4. $\mu_2 > \mu_3 > \mu_1$
5. $\mu_3 > \mu_1 > \mu_2$
6. $\mu_3 > \mu_2 > \mu_1$

H_u combines H_1, H_2, and H_c.

PMPs

Replacing H_u by H_c

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.04	1/5
H_2 : Gender Role Match	.0725	.012	5.85	.84	1/5
H_3 : Sex Mismatch	.0007	.012	.06	.00	1/5
H_4 : Gender Role Mismatch	.0001	.012	.01	.00	1/5
H_c :	.9200	.9500	.97	.12	1/5

where H_c denotes the complement H_1 through H_4, that is, "not one of these four hypotheses".

PMPs

Bayesian Error Probabilities

PMPs can be interpreted as Bayesian (conditional!) error probabilities, that is, the Bayesian counterparts of the Type I and Type II errors.

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.04	1/5
H_2 : Gender Role Match	.0725	.012	5.85	.81	1/5
H_3 : Sex Mismatch	.0007	.012	.06	.01	1/5
H_4 : Gender Role Mismatch	.0001	.012	.01	.00	1/5
H_u :				.14	1/5

PMPs

The Number of Hypotheses and PMPs

Look what happens if we compare many hypotheses, the PMPs become smaller and smaller, and thus the Bayesian error probabilities become larger and larger:

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.013	1/13
H_2 : Gender Role Match	.0725	.012	5.85	.270	1/13
H_3 : Sex Mismatch	.0007	.012	.06	.003	1/13
H_4 : Gender Role Mismatch	.0001	.012	.01	.000	1/13
H_5 : Lets try this one too	.0521	.012	2.61	.180	1/13
...					
H_{12} : Don't miss something	.0164	.012	1.36	.040	1/13
H_u :				.047	1/13

PMPs

The Number of Hypotheses and PMPs

The same results as two slides up are in fact obtained by assigning PMPs of 0 to each hypothesis that is NOT considered:

	f_m	c_m	BF_{mu}	PMP_m	PRI_m
H_1 : Sex Match	.0039	.012	.32	.04	1/5
H_2 : Gender Role Match	.0725	.012	5.85	.81	1/5
H_3 : Sex Mismatch	.0007	.012	.06	.01	1/5
H_4 : Gender Role Mismatch	.0001	.012	.01	.00	1/5
H_5 : Lets try this one too	.0521	.012	2.61	.18	0
...					
H_{12} : Don't miss something	.0164	.012	1.36	.04	0
H_u :				.14	1/5

Bayes Factor (BF)

comparing two informative hypotheses

The BF quantifies the relative support in the data for two hypotheses.

$$BF_{12} = \frac{BF_{1u}}{BF_{2u}} = \frac{f_1/f_2}{c_1/c_2}$$

using

$$BF_{mu} = \frac{f_m/f_u}{c_m/c_u} = \frac{f_m}{c_m},$$

since $f_u = 1$ and $c_u = 1$.

A Closer Look at the Bayes Factor

Three Simple Hypotheses

Consider the hypotheses:

$$H_1 : \mu_1 \approx \mu_2, \text{ that is, } |\mu_1 - \mu_2| < .1$$

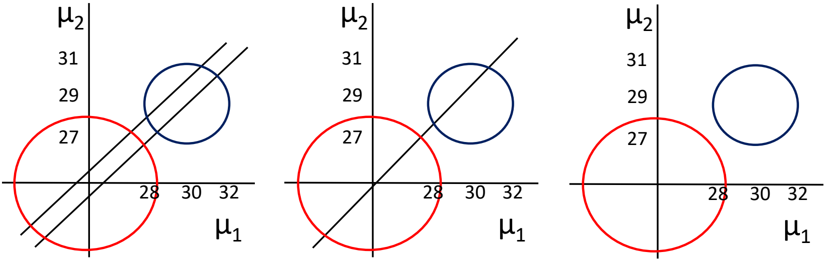
$$H_2 : \mu_1 > \mu_2$$

$$H_3 : \mu_1, \mu_2$$

A Closer Look at the Bayes Factor

Prior Distribution (c), Posterior Distribution (f), and Hypotheses

H_1: $\mu_1 \approx \mu_2$ H_2: $\mu_1 > \mu_2$ H_u: μ_1 , μ_2



$BF_{1u} = f_1/c_1 = .25/.05 = 5$ $BF_{2u} = f_2/c_2 = .75/.5 = 1.5$

$BF_{12} = 5/1.5 = 3.33$

A Closer Look at the Bayes Factor

Fit and Complexity

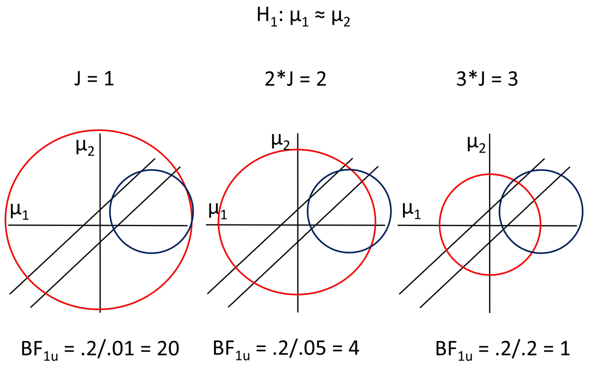
1. The **fit** of a hypothesis is the proportion of the **posterior** distribution in agreement with the hypothesis.
   Note: posterior = likelihood $\times$ prior.

2. The **complexity** of a hypothesis is the proportion of the **prior** distribution in agreement with the hypothesis.

A Closer Look at the Bayes Factor

Posterior Distribution (f), Prior Distribution (c), and Hypotheses
Prior Sensitivity for = Constrained Hypotheses

Here, used bain: J = 1.
Sensitivity check: J, 2*J, and 3*J (i.e., fraction = 1, 2, and 3).

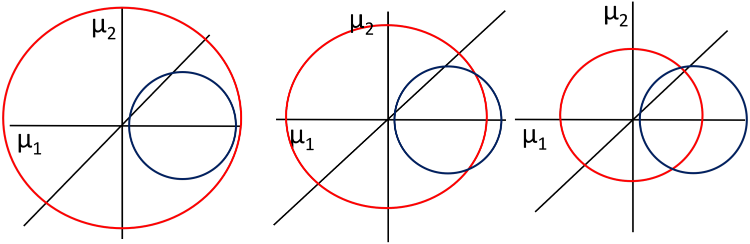


A Closer Look at the Bayes Factor

Posterior Distribution (f), Prior Distribution (c), and Hypotheses
Prior In-Sensitivity for >< Constrained Hypotheses

$H_2: \mu_1 > \mu_2$

J = 1 | 2*J = 2 | 3*J = 3



$BF_{2u} = .9/.5 = 1.8$ | $BF_{2u} = .9/.5 = 1.8$ | $BF_{2u} = .9/.5 = 1.8$